ME 460 Project - SOLAR ENERGY TECHNOLOGY

SOLAR WHEEL CHAIR FOR PHYSICALLY DISABLED

INTRODUCTION

1. Mobility assistance is vital for physically challenged individuals. Electric wheelchairs offer independence but demand frequent charging from grid electricity.

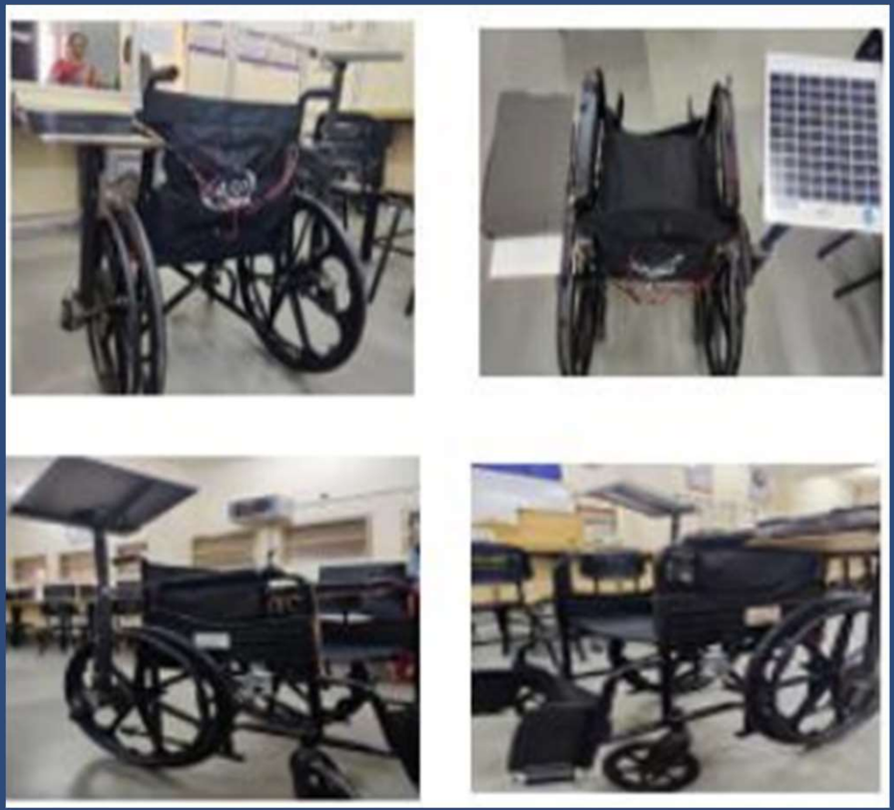
2. Solar energy is a clean, renewable source capable of continuously powering mobility devices without utility dependency.

3. This project demonstrates a solar-powered wheelchair controlled by a Joystick module, integrating photovoltaic charging with embedded motor control.

THE MAIN OBJECTIVES OF THIS PROJECT ARE:

- ❖ To design a solar powered wheelchair.
- ❖ To utilize solar energy for charging the battery.
- ❖ To implement Joystick-based control for directional movement.
- ❖ To demonstrate a practical application of solar energy technology.

SOLAR PANEL



WHY WE CHOOSE SOLAR TECHNOLOGY FOR WHEEL CHAIR

♻️ **Renewable Energy Source**

Sunlight is inexhaustible — ideal for continuous outdoor use.

💰 **Cost-effective & Eco-friendly**

Eliminates recurring electricity bills; zero carbon emissions.

🧭 **Supports Independent Mobility**

Users can travel without worrying about plug-in charging.

⚡ **Useful in Areas with Limited Electricity**

Critical for rural regions with unreliable grid access.

🔧 **Low Maintenance**

Solar panels have no moving parts; minimal upkeep required.

⚙ REQUIRED COMPONENTS & SPECIFICATIONS

Solar Panel

12V, 20W | 470×350×25 mm

Converts sunlight → DC electricity via photovoltaic effect. Peak power: 20W at standard conditions.

Solar Charge Controller

PWM, 10A | 115×70×35 mm

Regulates voltage & current from panel to battery; prevents overcharge & deep discharge.

Rechargeable Battery

12V, 7Ah SLA | 151×65×94 mm

Stores 84Wh of energy. Supplies sustained power to motors and control electronics.

Arduino Uno

ATmega328P, 5V | 68.6×53.4 mm

Reads joystick analog signals; outputs PWM to motor driver. 14 digital I/O, 6 analog inputs.

XL4015 Buck Converter

5A, 75W max | 54×23×15 mm

Steps down 12V battery to 5V for Arduino & logic circuits. Efficiency ≈ 96%.

Joystick Module

2-axis analog | 40×27×32 mm

X/Y potentiometers output 0–5V analog signals. Controls forward/back/left/right movement.

BTS7960 Motor Driver

43A peak, 5–27V | 55×55×12 mm

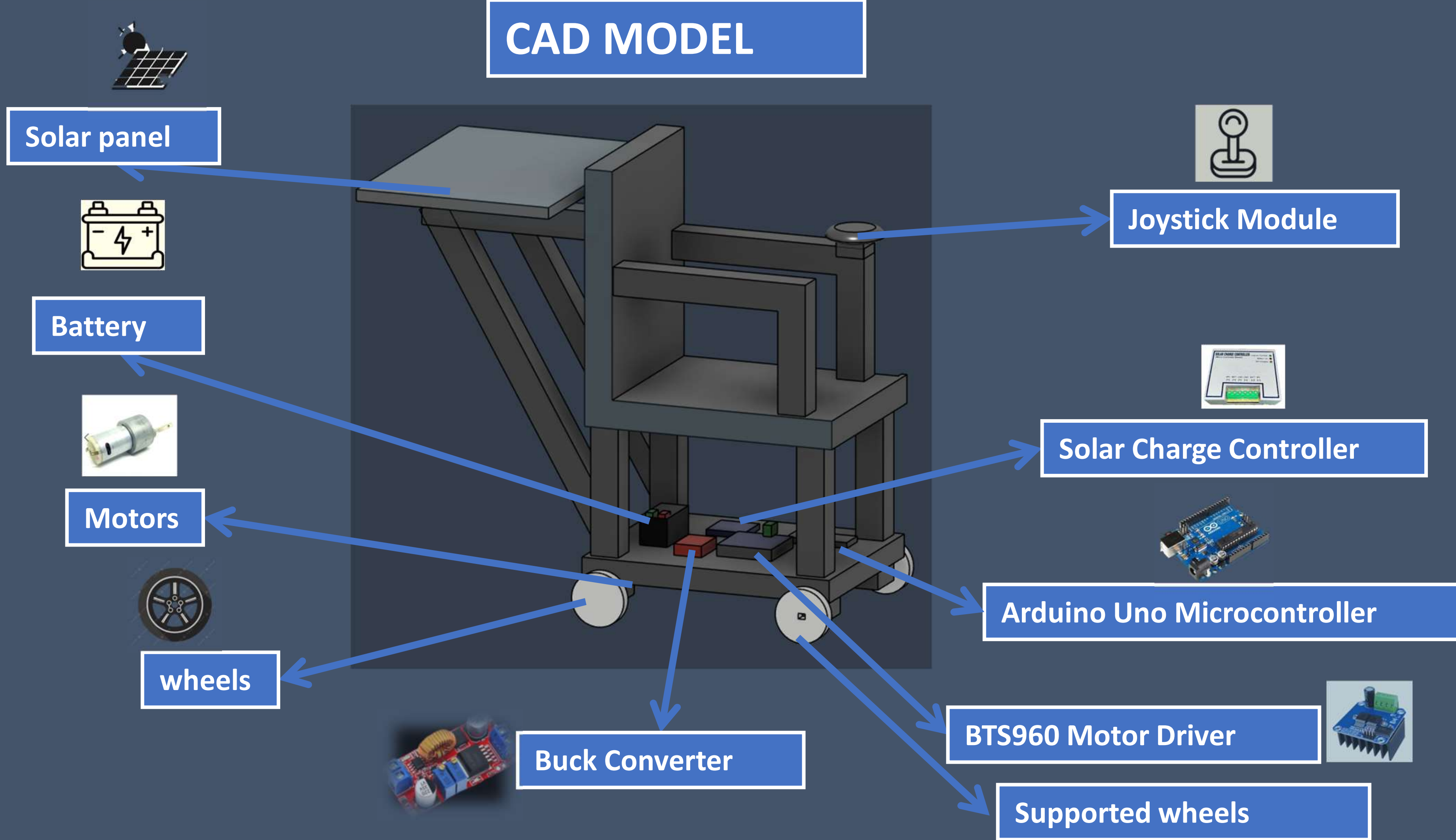
H-bridge driver for high-current DC motors. PWM frequency up to 25 kHz. Built-in protection.

DC Gear Motors

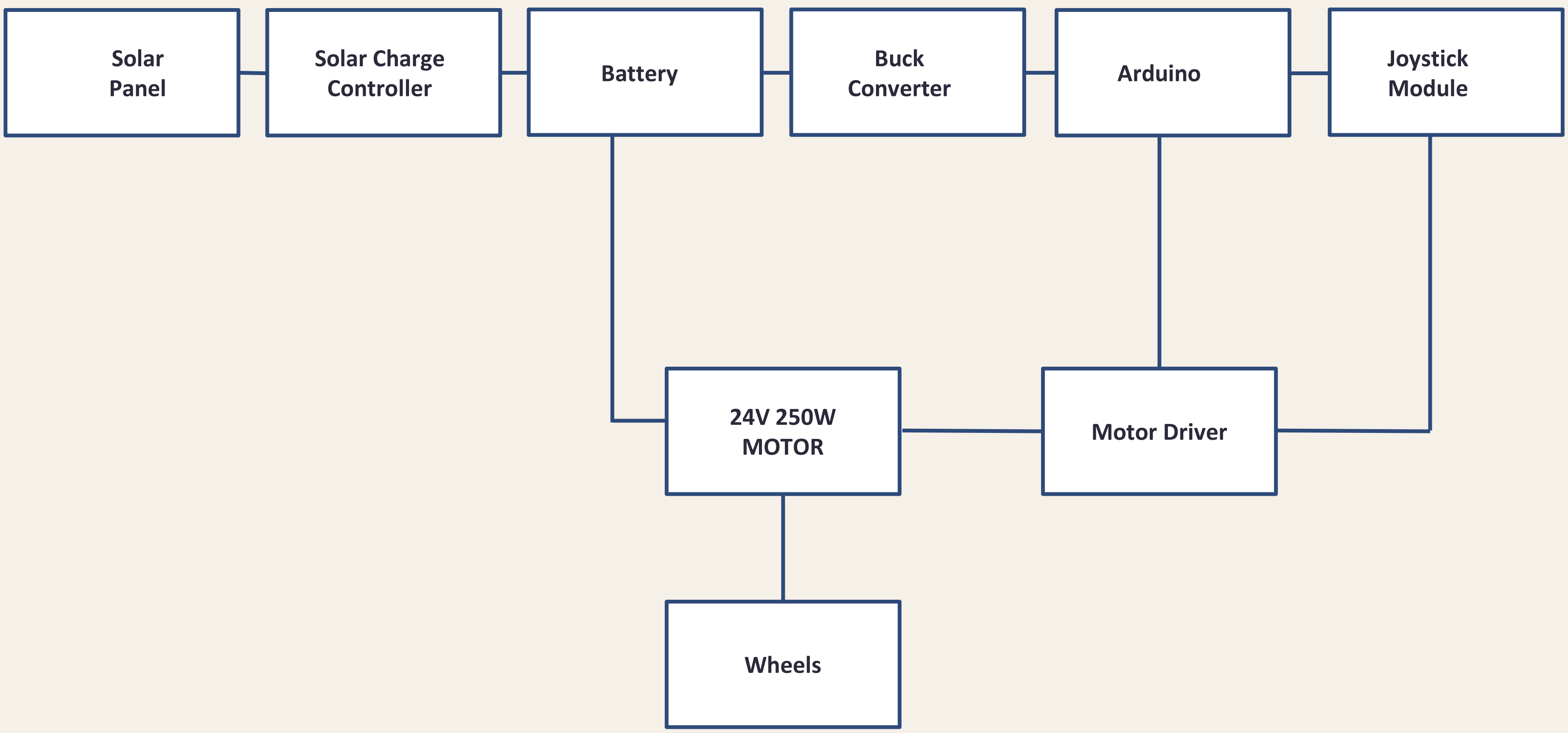
12V, 100–200 RPM | ø37×68 mm

High-torque geared motors provide force to drive wheelchair wheels on varied terrain.

CAD MODEL


Solar panel
Battery
Motors
wheels
Buck Converter
Joystick Module
Solar Charge Controller
Arduino Uno Microcontroller
BTS960 Motor Driver
Supported wheels

FLOW CHART



Solar Panel
Solar Charge Controller
Battery
Buck Converter
Arduino
Joystick Module
24V 250W MOTOR
Motor Driver
Wheels

WORKING PRINCIPLE

The system works using two main subsystems:

- **Solar Energy System**
- **Control System**

Working Process:

❖ The solar panel converts sunlight into DC electricity via the photovoltaic effect.

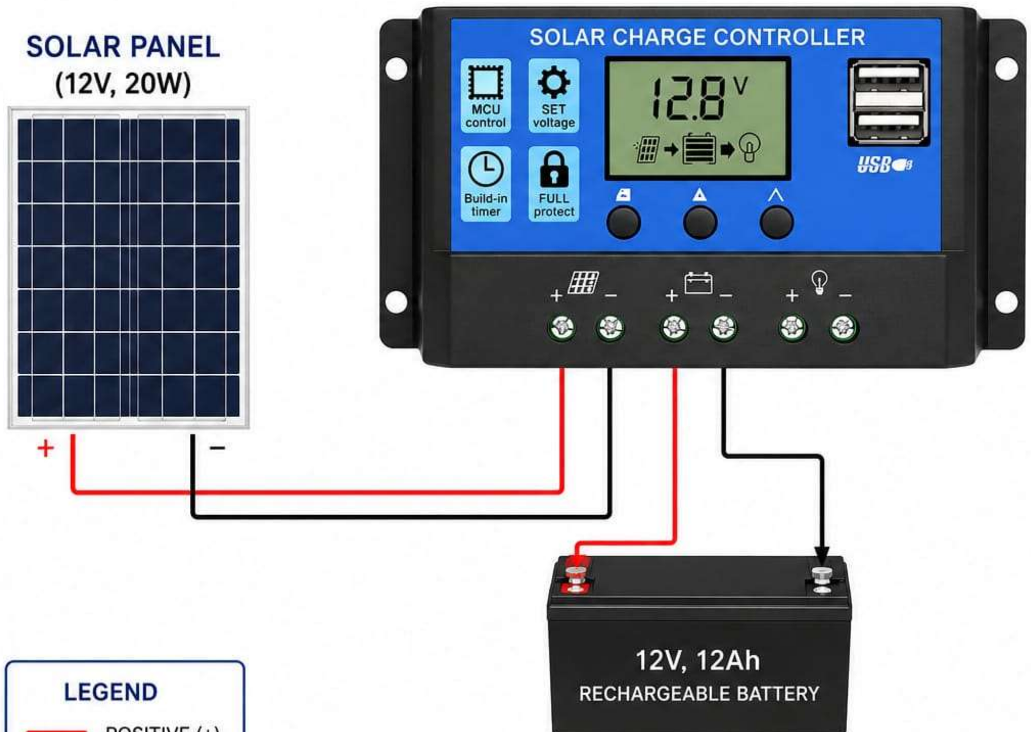
❖ The charge controller regulates voltage & current to ensure safe battery charging.

❖ The 12V 7Ah rechargeable battery stores all harvested solar energy.

❖ Stored energy powers the motor driver, DC motors, and control electronics.

Energy Flow: Sunlight → Solar Panel → Charge Controller → Battery

SOLAR CHARGING SYSTEM CIRCUIT DIAGRAM



CONNECTION STEPS

1. Connect battery to controller first.
2. Connect solar panel to controller.
3. Controller will regulate the charging current and protect the battery.

LEGEND

- POSITIVE (+)
- NEGATIVE (–)

NOTE

- The solar charge controller protects the battery from overcharging, over-discharge and reverse polarity.

Control System

The wheelchair is controlled by a Joystick module interfaced with Arduino Uno.

❖ User moves joystick in desired direction (forward, backward, left, right).

❖ The Joystick Module detects movement and generates corresponding 0–5V analog signals.

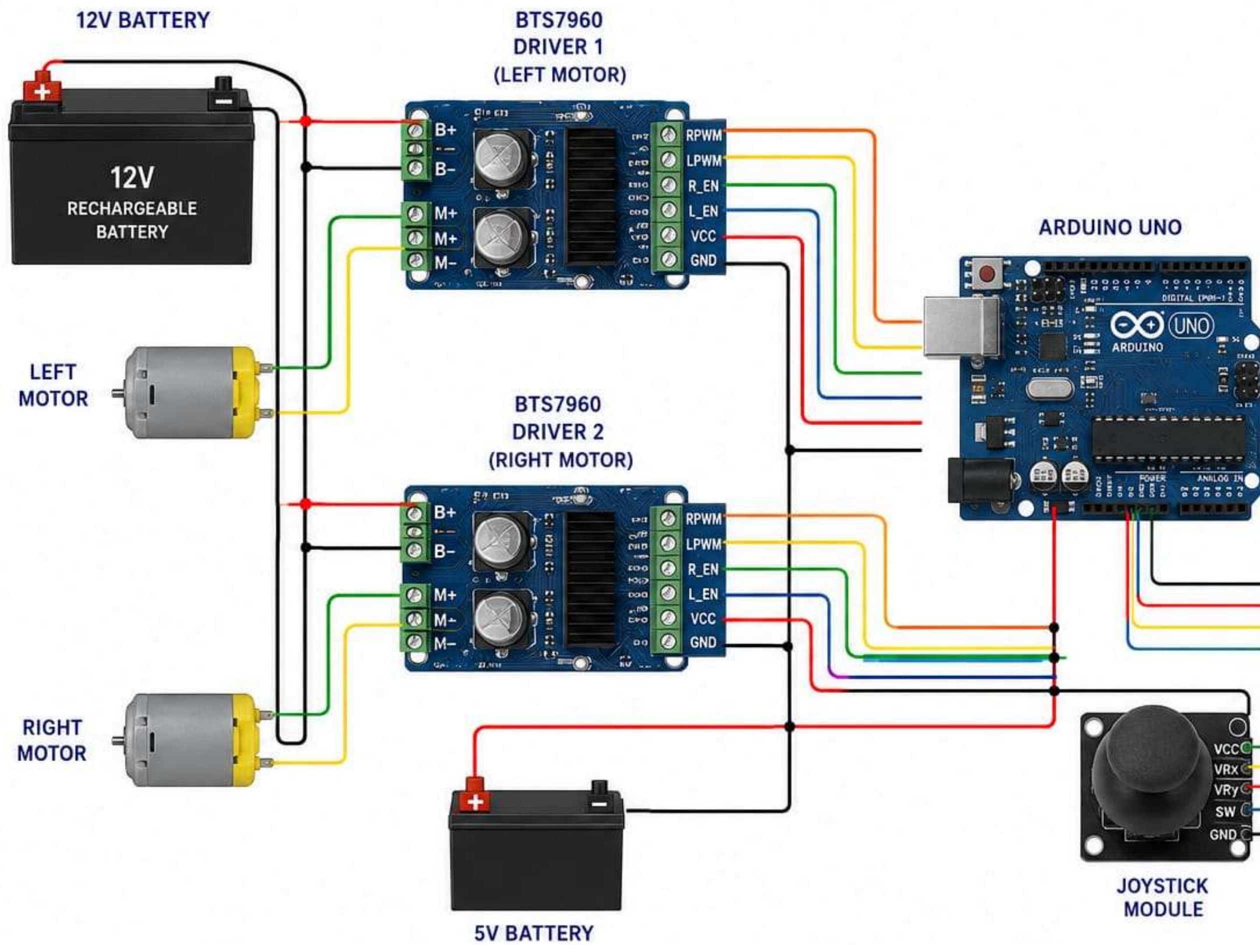
❖ Arduino Uno reads X/Y axis values via analog input pins A0 and A1.

❖ Arduino processes input and sends PWM signals to the BTS7960 Motor Driver.

❖ Motor Driver supplies high current from battery to drive DC motors in correct direction.

Control Flow: Joystick → Arduino → Buck Converter → Motor Driver → Motors

CONTROL SYSTEM CIRCUIT DIAGRAM
(JOYSTICK BASED)



1. BATTERY CONNECTIONS
Battery + → B+ of Driver1 & Driver2
Battery – → B- of Driver1 & Driver2
Battery – → Arduino GND
Common ground is REQUIRED.

2. MOTOR CONNECTIONS
Motor 1 → Driver 1 output (M+ , M-)
Motor 2 → Driver 2 output (M+ , M-)

3. JOYSTICK CONNECTIONS

Joystick Pin	Arduino
VCC	5V
GND	GND
VRX	A0
VRY	A1
SW	Optional

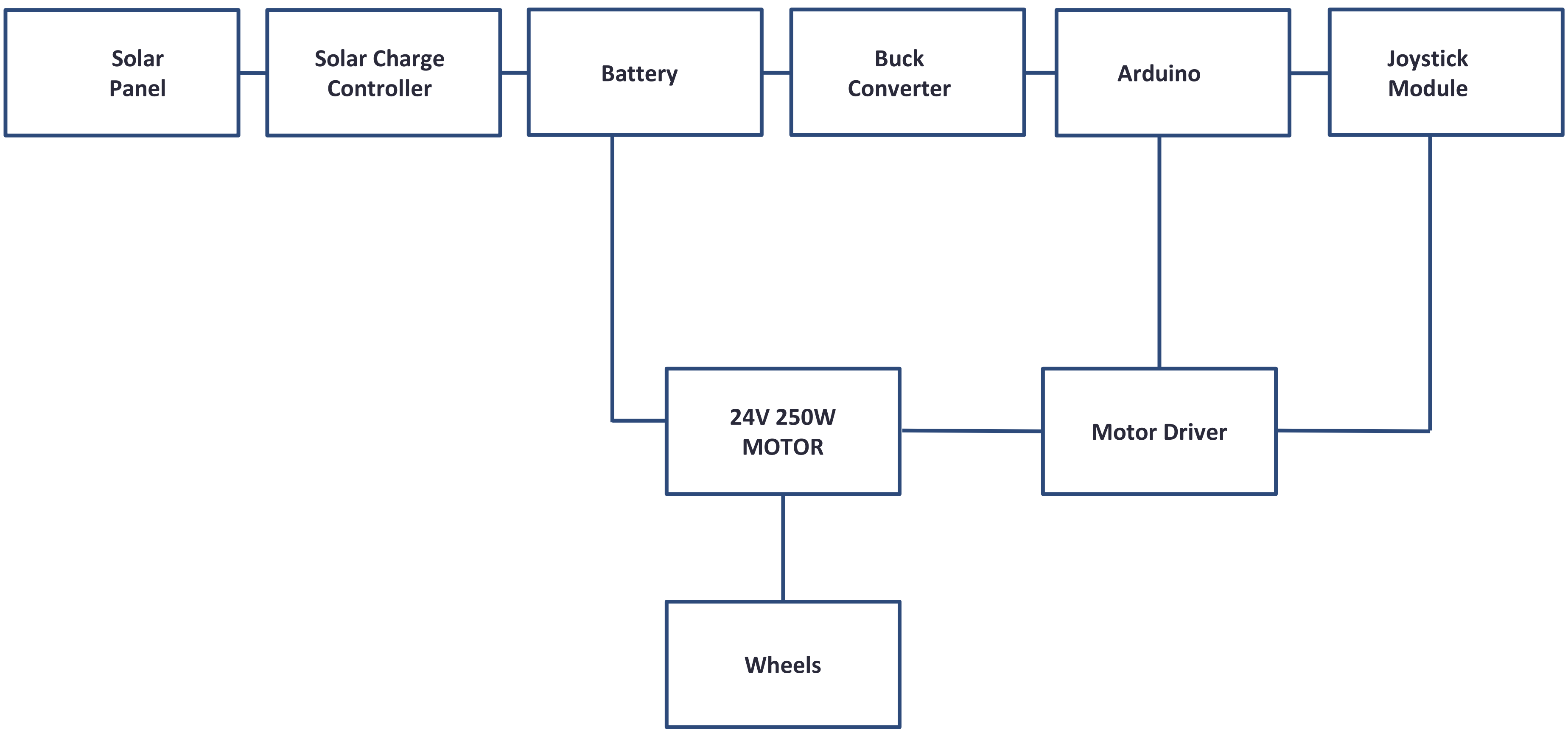
4. BTS7960 DRIVER 1 (LEFT MOTOR)

BTS7960	Arduino
RPWM	D3
LPWM	D5
R_EN	D2
L_EN	D4
VCC	5V
GND	GND

5. BTS7960 DRIVER 2 (RIGHT MOTOR)

BTS7960	Arduino
RPWM	D9
LPWM	D10
R_EN	D7
L_EN	D8
VCC	5V
GND	GND

FLOW CHART

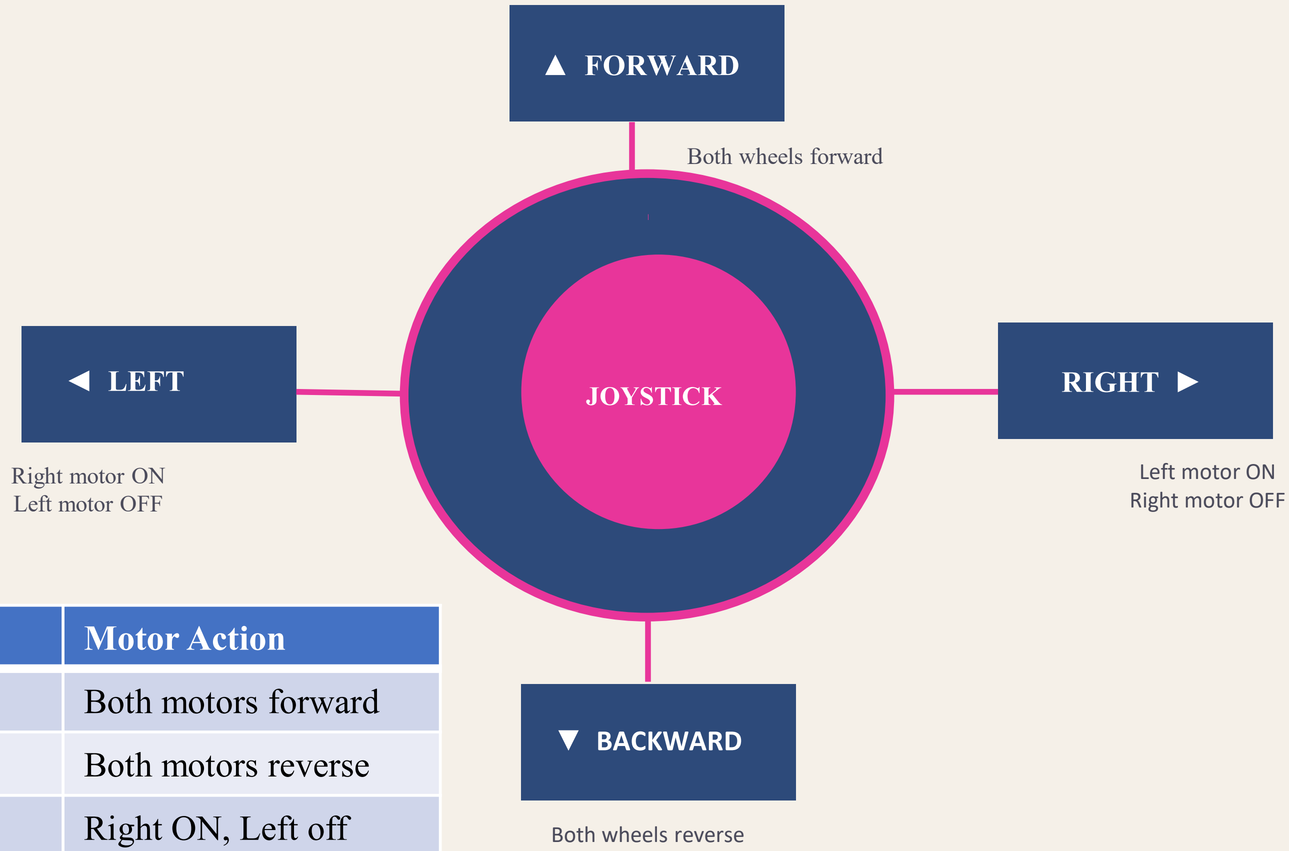


Solar Panel
Solar Charge Controller
Battery
Buck Converter
Arduino
Joystick Module
24V 250W MOTOR
Motor Driver
Wheels

Programming

```
// ===========================
// JOYSTICK + 2 BTS7960 + 2 MOTORS
// ============================

// -------- MOTOR 1 --------
int R_IS1  = 1;
int R_EN1  = 2;
int R_PWM1 = 3;
int L_IS1  = 4;
int L_EN1  = 5;
int L_PWM1 = 6;
// -------- MOTOR 2 --------
int R_IS2  = 11;
int R_EN2  = 7;
int R_PWM2 = 9;
int L_IS2  = 12;
int L_EN2  = 8;
int L_PWM2 = 10;

// -------- JOYSTICK --------
int VRX = A0;
int VRY = A1;

int xValue = 0;

void setup() {

  // ========================
  // MOTOR 1 PINS
  // ========================
  pinMode(R_IS1, OUTPUT);
  pinMode(R_EN1, OUTPUT);
  pinMode(R_PWM1, OUTPUT);
  pinMode(L_IS1, OUTPUT);
  pinMode(L_EN1, OUTPUT);
  pinMode(L_PWM1, OUTPUT);

  // ========================
  // MOTOR 2 PINS
  // ========================
  pinMode(R_IS2, OUTPUT);
  pinMode(R_EN2, OUTPUT);
  pinMode(R_PWM2, OUTPUT);
  pinMode(L_IS2, OUTPUT);
  pinMode(L_EN2, OUTPUT);
  pinMode(L_PWM2, OUTPUT);

  // ========================
 // ENABLE BOTH DRIVERS
  // ========================
  digitalWrite(R_IS1, LOW);
  digitalWrite(L_IS1, LOW);
  digitalWrite(R_EN1, HIGH);
  digitalWrite(L_EN1, HIGH);

  digitalWrite(R_IS2, LOW);
  digitalWrite(L_IS2, LOW);
  digitalWrite(R_EN2, HIGH);
  digitalWrite(L_EN2, HIGH);

  Serial.begin(9600);
}

// ========================
  // ENABLE BOTH DRIVERS
  // ========================
  digitalWrite(R_IS1, LOW);
  digitalWrite(L_IS1, LOW);
  digitalWrite(R_EN1, HIGH);
  digitalWrite(L_EN1, HIGH);

  digitalWrite(R_IS2, LOW);
  digitalWrite(L_IS2, LOW);
  digitalWrite(R_EN2, HIGH);
  digitalWrite(L_EN2, HIGH);

  Serial.begin(9600);
}



// ===================================
  // LEFT TURN
  // ===================================
  else if(xValue < 400){

    analogWrite(R_PWM1, 0);
    analogWrite(L_PWM1, 0);

    analogWrite(R_PWM2, 200);
    analogWrite(L_PWM2, 0);
  }

  // ===================================
 // RIGHT TURN
  // ===================================
  else if(xValue > 600){

    analogWrite(R_PWM1, 200);
    analogWrite(L_PWM1, 0);

    analogWrite(R_PWM2, 0);
    analogWrite(L_PWM2, 0);
  }
  // STOP
  // ===================================
  else{

    analogWrite(R_PWM1, 0);
    analogWrite(L_PWM1, 0);

    analogWrite(R_PWM2, 0);
    analogWrite(L_PWM2, 0);
  }

  delay(50);
}

// ===================================
  // LEFT TURN
  // ===================================
  else if(xValue < 400){

    analogWrite(R_PWM1, 0);
    analogWrite(L_PWM1, 0);

    analogWrite(R_PWM2, 200);
    analogWrite(L_PWM2, 0);
  }

  // ===================================
  // RIGHT TURN
  // ===================================
  else if(xValue > 600){

    analogWrite(R_PWM1, 200);
    analogWrite(L_PWM1, 0);

    analogWrite(R_PWM2, 0);
    analogWrite(L_PWM2, 0);
  }
  // STOP
  //
===================================
==
  else{

    analogWrite(R_PWM1, 0);
    analogWrite(L_PWM1, 0);

    analogWrite(R_PWM2, 0);
    analogWrite(L_PWM2, 0);
  }

  delay(50);
}
```

JOYSTICK MOVEMENT CONTROL



Direction	Motor Action
forward	Both motors forward
backward	Both motors reverse
Left	Right ON, Left off
Right	Left on , Right off
stop	Both motors stop

Calculations

1. Battery Energy Calculation
Battery Voltage = 12V
Battery Capacity = 7Ah

Formula:

$$\text{Battery } Energy\ (Wh) = V \times Ah$$

Calculation:

$$12 \times 7 = 84\ Wh$$

Result:
The battery stores approximately **84 Wh** of electrical energy.

2. Motor Power Calculation
Number of Motors = 2
Motor Voltage = 12V
Current drawn by each motor ≈ 2A
Total Current:

$$I_{total} = 2A + 2A = 4A$$

Formula:

$$P = VI$$

Calculation:

$$P = 12 \times 4 = 48\ W$$

Result:
The total power consumed by the motor system is approximately **48 W**.

3. Battery Backup Time Calculation
Battery Energy = 84 Wh
Total Load Power = 48 W
Formula:

$$\text{Backup } Time = \frac{Battery\ Energy\ (Wh)}{Load\ Power\ (W)}$$

Calculation:

$$\frac{84}{48} = 1.75\ hours$$

Result:
The wheelchair can operate continuously for approximately **1.7 hours** under normal load conditions.

4. Solar Panel Current Output Calculation
Solar Panel Power = 20W
Solar Panel Voltage = 12V
Formula:

$$I = \frac{P}{V}$$

Calculation:

$$I = \frac{20}{12} = 1.67A$$

Result:
The solar panel provides approximately **1.67 A** output current.

5. Battery Charging Time Calculation
Battery Capacity = 84 Wh
Solar Panel Power = 20W
Formula:

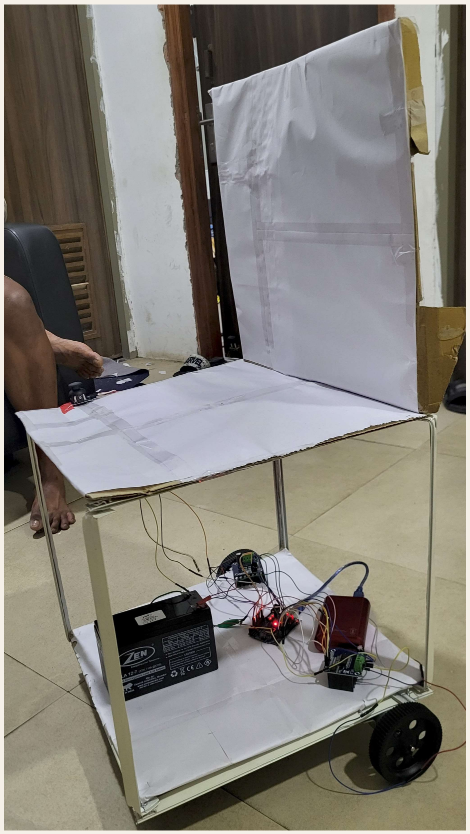
$$\text{Charging } Time = \frac{Battery\ Capacity\ (Wh)}{Solar\ Panel\ Power\ (W)}$$

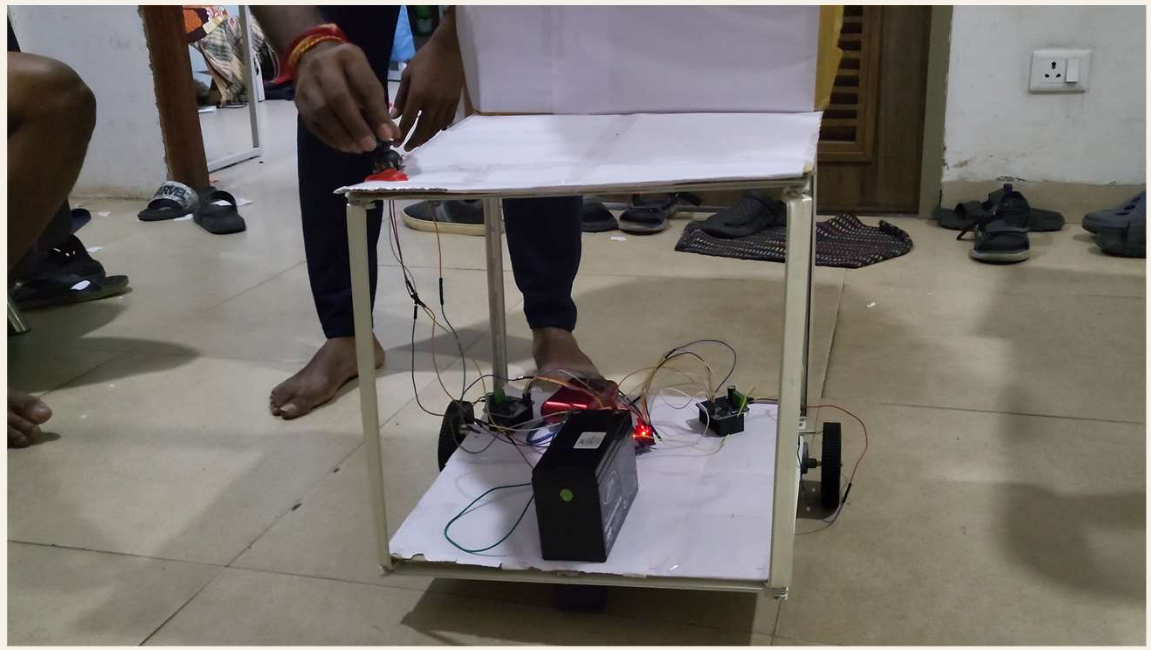
Calculation:

$$\frac{84}{20} = 4.2\ hours$$

Result:
The battery requires approximately **4–5 hours** for complete charging under ideal sunlight conditions.

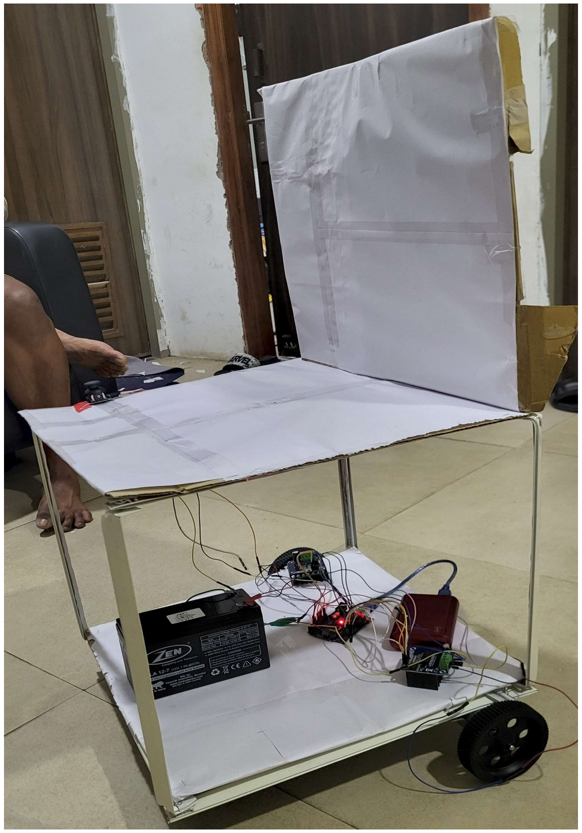
Designed Wheel Chair





ADVANTAGES

- Uses renewable solar energy — self-charging while outdoors.
- Environment-friendly system — zero carbon footprint during operation.
- Reduces dependency on grid electricity — suitable for remote areas.
- Joystick control — intuitive, safe, and precise directional movement.
- Suitable for outdoor mobility assistance in varied terrain.



BUDGET OF THE COMPONENTS

Component	Specification	Cost (₹)
Solar Panel	12V, 20W	999
Solar Charge Controller	PWM 20A	505
Battery	12V 7Ah SLA	1350
Arduino Uno	ATmega328P (available)	400
Joystick Module	2-axis analog	135
Motors & Wheels	12V DC Gear Motors ×2	1186
BTS7960 Motor Driver	43A H-Bridge ×2	1190
XL4015 Buck Converter	5A, 75W	300
TOTAL		**₹ 6,065**

CONCLUSION

This project demonstrates a solar-powered wheelchair with joystick-based control. Solar energy charges the battery and powers the entire system, reducing dependence on conventional energy sources. The joystick allows users to easily and safely control wheelchair movement in four directions: forward, backward, left, and right. The integration of renewable energy with embedded electronics Arduino Uno, BTS7960 motor driver, and DC gear motors provides a smart, reliable, and sustainable mobility solution for physically disabled individuals.

THANK YOU

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